Project Context:

We are developing a tool that takes a gene as input and identifies which miRNAs may regulate that gene. For example, the user could enter PTEN and the program would return a ranked list of miRNAs predicted to target PTEN, along with scores indicating the strength of the predictions. The long-term goal is to combine predictions from multiple miRNA target-prediction algorithms into an integrated/consensus score. The reason for using multiple algorithms is that different algorithms use different biological features, datasets, mathematical approaches, and assumptions. An interaction predicted strongly by several different approaches may be more compelling than an interaction predicted by only one algorithm.

miRNAs (microRNAs) are small non-coding RNA molecules that regulate gene expression. A miRNA can bind to a complementary sequence in an mRNA and cause reduced protein production and/or mRNA degradation. The interaction is usually not perfectly complementary, which makes computational prediction difficult. The seed region of a miRNA, generally nucleotides 2–7/8 from the miRNA's 5' end, is especially important for target recognition.

An miRNA target is an mRNA/gene that can be regulated by a particular miRNA. More specifically, prediction algorithms often predict a target site within an mRNA, commonly in the 3′ untranslated region (3′ UTR), although some algorithms also consider coding sequences (CDS). A simplified relationship is: miRNA → target site → mRNA/gene.

Target-prediction algorithms are necessary because there are thousands of miRNAs and tens of thousands of genes, creating an enormous number of possible interactions. Experimentally testing every possible interaction would be extremely expensive and time-consuming. Computational algorithms therefore prioritize likely interactions by looking at features such as seed sequence matching, evolutionary conservation, binding/free energy, RNA accessibility, miRNA–mRNA pairing, target-site abundance, local sequence composition, transcript position, 3′ compensatory pairing, and other sequence/context features. Different algorithms use different combinations of these features.

Examples of miRNA target-prediction resources include TargetScan, DIANA-microT-CDS, miRDB, and mirDIP. Examples of experimentally supported interaction databases include miRTarBase and DIANA-TarBase.

IMPORTANT DISTINCTION: TargetScan, DIANA, miRDB, and similar resources are used for computational prediction. miRTarBase and TarBase are experimentally supported miRNA–target interaction databases. They are not prediction algorithms. They collect and curate evidence from laboratory experiments that support miRNA–target interactions. Examples of experimental evidence can include reporter assays, Western blot/protein measurements, qPCR, CLIP-based experiments, and knockdown or overexpression experiments.

The overall project architecture is currently: Gene → TargetScan → DIANA → mirDIP/other prediction resources → integrated score → experimental validation/reference using miRTarBase and TarBase. The current R program is being built modularly so that each prediction algorithm

can eventually produce its own score, after which the scores can be compared and potentially integrated.

The purpose of this assignment is to research and compare different miRNA target-prediction algorithms/resources so that we understand how they actually work before deciding how to integrate their scores. We are NOT simply comparing websites. We need to understand the computational logic behind each method.

For each algorithm/resource, research the following:

1. What is it?
- Who developed it?
- When was it introduced?
- What is the current/relevant version?
- What problem does it solve?
2. How does it work?
   Explain the process from input to final prediction. Ideally describe:
   miRNA/mRNA information → candidate target sites → biological/computational features → mathematical/statistical/machine-learning model → prediction score → ranking/output.
3. What biological/computational features does it use?
   Look specifically for:
- Seed matching
- 8mer/7mer/6mer site types
- Evolutionary conservation
- Binding/free energy
- RNA/site accessibility
- Local AU content
- Target-site abundance
- Position within transcript
- 3′ compensatory pairing
- G:U pairing
- Machine learning
- Other sequence/context features

Do not assume that an algorithm uses a feature. Verify it from documentation or papers.

4. What does its score mean?
   This is extremely important for our project. Find:
- The name of the score
- The numerical range
- Whether higher or lower values indicate stronger predictions
- Whether it is a probability, confidence score, ranking score, energy score, etc.
- Whether the score has an absolute interpretation across genes
- Whether the score is already normalized

- Whether there is a recommended cutoff/threshold
- Whether multiple site scores are aggregated into a gene–miRNA score

5. Advantages:
   Research the actual strengths of the method. Examples could include conservation, high coverage, machine learning, ability to detect non-conserved targets, use of multiple biological features, ability to consider CDS sites, or integration of multiple prediction methods. Do not simply assume these are advantages; support them with sources.
6. Disadvantages:
   Research limitations such as false positives, false negatives, dependence on conservation, dependence on training data, inability to account for tissue-specific expression, limitations in modeling RNA structure, algorithmic assumptions, or correlated evidence. Again, verify these rather than simply listing generic limitations.
7. When would you use it?
   Give a realistic research scenario where this algorithm/resource would be useful and explain why.

ALGORITHMS/RESOURCES TO RESEARCH:

TARGETSCAN:
TargetScan is one of the major miRNA target-prediction methods. It searches for miRNA recognition sites based heavily on seed-region matching and includes canonical site types such as 8mer, 7mer-m8, 7mer-A1, and 6mer. It also considers evolutionary conservation and other sequence/context features.

Important TargetScan concepts to research in depth:

- Seed matching
- 8mer, 7mer-m8, 7mer-A1, and 6mer sites
- Context++ score
- Weighted Context++ score
- Aggregate/cumulative scores
- PCT (Probability of Conserved Targeting)
- Evolutionary conservation
- How TargetScan ranks predictions
- What changed in current TargetScan versions
- How the different TargetScan scores should be interpreted

TargetScan's Context++ score combines contributions from multiple biological/context features, while PCT provides information related to preferential evolutionary conservation. These should be treated as different types of evidence rather than assuming they measure the same thing.

For TargetScan, we especially want to understand the actual mathematical logic behind Context++ and how individual site-level scores are combined into gene–miRNA-level predictions.

DIANA-microT-CDS:
DIANA-microT-CDS is another major miRNA target-prediction method. It considers miRNA recognition elements in both the 3′ UTR and coding sequence (CDS). It uses multiple biological/computational features, including seed matching, conservation, free energy, site accessibility, target-site abundance, machine learning, 3′ compensatory pairing, G:U pairing, and local AU content.

Research:

- What microT-CDS stands for and what it predicts
- How candidate sites are identified
- How seed matching is used
- How conservation is used
- How free energy is used
- How site accessibility is modeled
- How CDS targeting differs from 3′ UTR targeting
- What machine-learning component is used
- What its prediction score represents
- Its score range and cutoff
- Advantages/disadvantages compared with TargetScan

mirDIP:
mirDIP is different from TargetScan and DIANA because it is an integrative prediction resource rather than simply one individual prediction algorithm. It combines predictions from multiple independent miRNA target-prediction resources and produces an integrated score for miRNA–gene interactions.

Conceptually:
TargetScan + DIANA + miRDB + PITA + RNAhybrid + other prediction resources → mirDIP → integrated prediction score.

Research:

- Which prediction resources mirDIP integrates
- How the individual predictions are combined
- How the mirDIP integrated score is calculated
- What the integrated score means
- Its score range and confidence categories
- How mirDIP evaluates predictions against experimentally supported interactions
- Advantages of integrating multiple algorithms
- Disadvantages of relying on an integrated resource
- Whether combining multiple prediction methods creates correlated evidence or other biases

Important: Do NOT describe mirDIP as simply "another algorithm like TargetScan." It is an integrative database/resource that combines predictions from multiple sources.

miRDB:
Also research miRDB if there is room in the presentation. miRDB provides predictions from its MirTarget machine-learning system. Research how its machine-learning model works, what training data it uses, what features it considers, what its score means, its range/cutoff, and its strengths and weaknesses. This is useful because it provides a machine-learning-based contrast to methods such as TargetScan and DIANA.

VALIDATION DATABASES:

miRTarBase:
miRTarBase is a database of experimentally supported miRNA–target interactions. It can be used as a reference for determining whether a computationally predicted interaction has experimental evidence.

DIANA-TarBase:
TarBase is another database of experimentally supported miRNA–target interactions. It contains curated experimental evidence for miRNA–target relationships.

For our project, these databases may eventually be used as reference/validation data. We are NOT using them as prediction algorithms. The distinction is:

Prediction algorithms:
"Based on computational evidence, could this miRNA regulate this gene?"

Experimental validation databases:
"Has this miRNA–gene interaction been supported by laboratory experiments?"

WHY WE ARE DOING THIS:

Our ultimate goal is to build a consensus miRNA target finder. The current version of the R program can obtain TargetScan predictions and calculate a TargetScan-based score. We plan to add DIANA and potentially mirDIP and other resources later.

The future architecture is approximately:

Gene
↓
TargetScan prediction + score
↓
DIANA prediction + score
↓
mirDIP/other prediction scores
↓

Integrated/consensus score
↓
Compare against experimentally supported interactions from miRTarBase/TarBase

However, we do NOT want to blindly average the scores. Before creating an integrated score, we need to understand what each score represents. For example, a TargetScan score may represent predicted targeting efficacy, PCT represents evolutionary conservation evidence, a DIANA score may represent that algorithm's prediction confidence, and a mirDIP score is already an integration of multiple prediction resources. These values may not be directly comparable.

Therefore, the main purpose of the research is to determine whether and how the scores from different algorithms can be normalized, compared, weighted, or otherwise integrated.

FINAL SLIDESHOW:

A possible structure is:

1. Why miRNA target prediction is difficult
2. What is an miRNA and what is an miRNA target?
3. Prediction vs. experimental validation
4. How miRNA target-prediction algorithms generally work
5. TargetScan
6. TargetScan scoring system
7. DIANA-microT-CDS
8. DIANA scoring system
9. miRDB
10. mirDIP
11. miRTarBase and TarBase
12. Side-by-side comparison
13. Why different algorithms produce different predictions
14. Advantages and disadvantages of each method
15. What this means for our consensus-scoring project

THE MOST IMPORTANT QUESTION THE SLIDESHOW SHOULD ANSWER:

Why can TargetScan, DIANA, miRDB, and mirDIP give different predictions for the same gene?

The answer should be based on differences in their:

- Candidate-site selection
- Seed-matching rules
- Conservation models
- Thermodynamic calculations
- Accessibility models
- Machine-learning models

- Training datasets
- Scoring systems
- Aggregation methods
- Underlying assumptions

For example, for a gene such as PTEN, TargetScan might strongly predict one set of miRNAs while DIANA predicts a somewhat different set. That does not automatically mean one is "wrong." The algorithms are evaluating the interaction using different evidence.

When researching, prioritize the original algorithm documentation, official websites, and original/relevant research papers. Tools4miRs (https://www.tools4mirs.org/software/target_prediction/) can be used as a starting directory to identify algorithms and the types of features they use, but the actual methodology and scoring details should be verified using the algorithm's documentation or scientific publications.

For every important claim in the slideshow, record the source so we can distinguish documented facts from our own interpretation.